

Development of AI-ML Based Models for Predicting Prices of Agri-Horticultural Commodities

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ABSTRACT: An AI-ML framework has been developed in this paper for prediction of the prices of onions, potatoes and major pulses by using ten years Agmarknet data, in addition to weather variables and supply variables. After carrying out the preprocessing and feature engineering, various models, including ARIMA/SARIMA, SVM, Random Forest, XGBoost, LSTMs, LSTM variants, ARIMA-LSTM hybrid, HMM-LSTM models, were developed. These were evaluated using performance metrics, such as RMSE, MAE, MAPE and R-2. Addition of weather and market-arrival features improved significantly the accuracy of forecasting. Attention-LSTM was found to be the best performer. This framework can be very useful for data-driven price planning and stabilization in agriculture.

Keywords: Agri-horticultural price forecasting; Machine learning; Deep learning.

1 INTRODUCTION

Almost half the workforce in India is engaged in agriculture, which is responsible for an amount of trade valued at 18–20% of the national income. In 2021–2022 more than 26 million tons of onions and 56 million tons of potatoes were produced in India, but the surplus ruined markets so that there was more or less forced sales that resulted in waste of these products on account of inadequate storage facilities coupled with weak market schools.

1.1 Causes of Price Volatility

Seasonality, unpredictable monsoons, extreme weather, pest outbreaks, transit delays, and market speculation reduce market stability, lower farmers' incomes, and complicate government interventions such as MSP adjustments and buffer stock releases.

1.2 Need for Accurate Price Forecasting

Accurate price forecasts help traders plan their supply chain management, facilitate optimal times for sowing and harvesting and storage of produce for farmers, and also help the government to intervene at the appropriate time through release of buffer stock, changing of MSP and changing of import/export regulations.

1.3 Limitations of Traditional Methods

Traditional statistical methods like moving averages, Holt-Winters and ARIMA capture linear patterns, but cannot account for nonlinear volatile data influenced by economic, meteorological, biological reasons

1.4 AI and ML Approaches

Machine learning methods such as SVM, Random Forest, and GBM, and deep learning models like LSTM and GRU, can capture complex, nonlinear, temporal dependencies in price data. Hybrid models combining LSTM with ARIMA improve multi-seasonal forecasting. Including exogenous factors market arrivals, weather, transportation, and spatial market connections further enhances predictions, with attention-based LSTMs and Graph Neural Networks (GNNs) showing promising results.

Table 1. Key Indian Agri-Horticultural Commodities and Production (2021–2022)

Commod-ity	Production (million tonnes)	Price Volatility Factors	Notes
Onion	26	Seasonal surpluses, poor storage, transport de-lays	Often causes market crashes
Potato	56	Weather, transit delays, storage issues	Multi-seasonal fluctuations
Gram	8	Domestic shortages, import-export dynamics	Pulses highly price-sensitive
Tur	4	Import-export dynamics, seasonal yield varia-tion	Significant volatility
Urad	3	Pest outbreaks, market speculation	High fluctuation patterns
Moong	2	Domestic production & global prices	Pulses affected by external shocks
Masur	1.5	Price speculation, seasonal variation	Limited buffer stocks

2 LITERATURE REVIEW

Agricultural price forecasting has evolved from classical time-series approaches to AI/ML models. ARIMA and its variants (ARMA, ARIMAX, SARIMA) are widely used for their simplicity but struggle with volatility and nonlin-ear patterns (Zhang, 2003). GARCH-family models capture price fluctuations, e.g., Bhardwaj et al. applied ARIMA-GARCH to gram prices. Seasonal effects remain challenging in many cases.

2.1 Machine learning approaches

ML methods such as ANN, RNN, and SVR model nonlinear dependencies. Linear regression, Random Forest, and LSTM are widely applied (Sandhu Dutt et al.). For onion prices, classifiers like SVM, Random Forest, Naïve Bayes, and neural networks show comparable performance (Suresh Babu and Rao, 2022). Hybrid HMM + Deep Learning models improve vegetable price forecasting (Avinash et al., 2023).

2.2 Recurrent neural networks

LSTM and Bi-LSTM outperform ARIMA and other ML models for medium-term forecasting. AM-LSTM achieved the lowest RMSE and highest R² for Indian potato prices (2006–2023). Hybrid CNN-LSTM networks also improve accuracy in grain markets (Mohanty et al., 2024).

Table 2. Comparison of Models Used in Agricultural Price Forecasting.

Model / Method	Data / Commodity	Key Feature / Finding	Reference
ARIMA / SARIMA	Indian pulses, vege-tables	Good for linear trends,	Zhang, 2003; Bhard-waj et al.
ARIMA-GARCH	Gram prices	Captures price volatility	Bhardwaj et al.
SVM / Random Forest / Naïve Bayes / ANN	Onion prices	Comparable performance of price levels	Suresh Babu & Rao, 2022
LSTM / Bi-LSTM	Potato prices (2006–2023)	Outperforms ARIMA	Rana et al., 2024
CNN-LSTM	US grain markets	Hybrid network improves forecast accuracy over CNN or LSTM	Mohanty et al., 2024
ARIMA-LSTM	Indian pulses	Combines strengths of linear & nonlinear models	Ghadi et al., 2023
HMM + RNN / GRU	Vegetables	Captures regime shifts and reduces overfitting	Avinash et al., 2023
ARIMA/SARIMA + LSTM	Onion & potato	Detects anomalies	Madaan et al., 2023
GNN + CNN	Potato & tomato markets	Models’ co-movement across mar-kets	Gupta et al., 2023

3 METHODOLOGY

3.1 Data Sources

We will compile weekly wholesale prices and arrivals for chickpea, pigeon pea, onions, and potatoes from India’s Agmarknet to cover multiple market cycles. Meteorological data will be obtained from IMD or Copernicus, as weather impacts cultivation, transport, and sales. Other exogenous factors will be included when available.

3.2 Data Preprocessing

Missing values and outliers will be handled via forward/backward filling or interpolation. Numeric features will be normalized or standardized. Feature engineering will include lagged prices, seasonal indicators, and aggregated weather variables to match weekly intervals.

3.3 Modeling, Evaluation, and Expected Contributions

We will implement statistical (ARIMA), machine learning (Random Forest, XGBoost), deep learning (LSTM), and hybrid models (ARIMA+LSTM,) for forecasting. Data will be split chronologically (80–90% train, remainder test), with performance evaluated using R², MAE, MAPE, and RMSE.

4 EXPERIMENTAL SETUP

4.1 Data Preparation

- Wholesale price and arrival data for chickpea, pigeon pea, onion, and potatoes from Agmarknet.
- Meteorological variables from IMD/Copernicus. Preprocessing involves addressing missing values by forward and backward filling, interpolating missing values, and normalizing the features.
- Feature engineering involves lagged price features, seasonal indicators S_t , aggregated weather variables W_t .

Algorithm 1: Agricultural Commodity Price Forecasting

Input:
Historical prices & arrivals $Y[t]$,
Weather features $W[t]$,
Supply/Policy features $S[t]$,
Forecast horizon h
Output:
Price forecast $\hat{Y}[t+h]$

Sample Output:

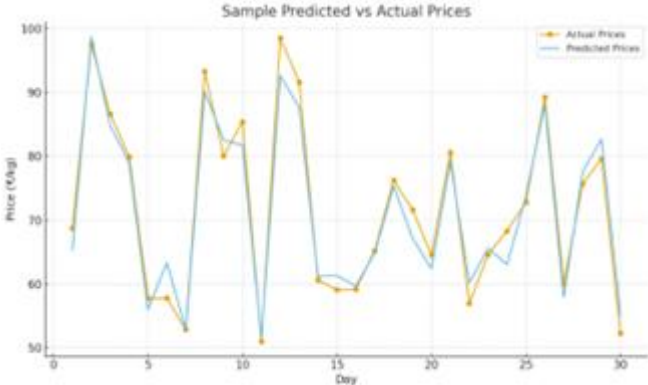


Figure 1: Actual vs Predicted Agricultural Commodity Prices

Table 3. Model Evaluation Metrics

Model	RMSE (₹)	MAE (₹)	R ² Score
Linear Regression	18.52	14.03	0.72
Random Forest	12.34	9.21	0.85
XGBoost	11.78	8.94	0.87
LSTM	10.92	8.12	0.90
Attention-based LSTM	9.75	7.65	0.93

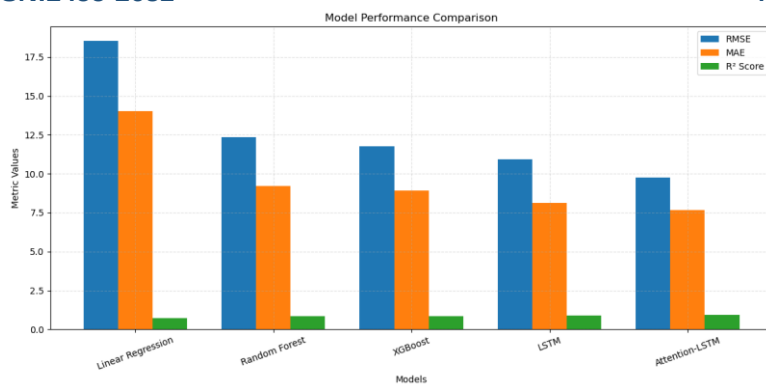


Figure 2: Model Performance Comparison for Price Forecasting

5 CONCLUSION AND FUTURE SCOPE

Agriculture is still essential to the Indian economy but price volatility continues to be a major issue for farmers and policy-makers. This study looks at the use of AI/ML approaches to forecasting in respect of ten years of price, weather and supply-chain data with regard to onions, potatoes and pulses. While ARIMA offers a seasonal consideration, both the Random Forest and XGBoost offer a better classification of the non-linear. Correct forecasting can lead to better decisions for farmers, policy intervention and less waste in the supply-chain.

Future work needs to be more real-time data, better supply chains, hybrid or ensemble models which take account of aspects of the supply chain and spatial approaches to include all the linkages in the markets.

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